Exact classical sampling of IBM's 70-qubit graph-state advantage experiment, and where the construction becomes hard

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Code, samples and raw outputs: https://github.com/chayworks/exact-graphstate-sampling (Apache-2.0).

Technical note, draft of 31 October 2026. Every number in this note is read from a committed artifact listed in the appendix, and every verification step has a negative control that fails as intended.

Abstract

In 2026 IBM and the University of Chicago posted a 70-qubit, depth-70 random-graph-state sampling experiment as a candidate for verifiable quantum advantage (Quantum Advantage Tracker #151), with a measured state fidelity of 0.37 and extrapolated classical costs of 10^{25} s (matrix product states) and 10^{7} s per sample (extended stabilizer simulation). We show that the instance is sampled **exactly from the ideal Born distribution** by a cut-rank dynamic program: **13.0 s per sample on one 16-core CPU node** (1,012 samples, 34 GB), and **18 s per sample on a laptop** with an 8.6 GB table. No rejection, no truncation, no GPU.

Three further results follow from having the exact probabilities in hand.

1. **IBM's own hardware shots, scored exactly.** The full-depth linear cross-entropy benchmark, which the instance README says "can only be extrapolated", is computed for every one of the 1,726 published shots: **XEB = 0.466 ± 0.031**, controls consistent with zero, and every per-shot probability agrees with an independently written program (tracker #260) to 10^{-5}. Two fidelity estimators built on the exact probabilities give 0.46 ± 0.04 and 0.48 [0.43, 0.54], above IBM's direct-fidelity estimate of 0.37 (95% lower bound 0.092) and much narrower.
2. **The frontier, measured rather than projected.** The width that governs the attack grows as 0.49·n − 2.9 for this construction at depth n, so memory doubles every two qubits. We sampled a 76-qubit, depth-76 instance exactly on the same 16-core node class (19 samples, 190 s each, 69 GB), with two samples reproduced to 10^{-7} by an independent engine. An 80-qubit instance is prepared (275 GB).
3. **A ceiling that holds for every graph.** Because a cut rank never exceeds min(|P|, n − |P|), *any* n-qubit graph state measured in *any* product basis is exactly samplable with 2^{⌊n/2⌋} table entries: 275 GB at 70 qubits, 2.2 TB at 76, 281 TB at 90, 9 PB at 100. No graph-state sampling experiment with product measurements can escape a single large-memory node below about 76 qubits, or a supercomputer below about 90, whatever graph is chosen. The random brickwork construction already sits within 3–4 of this ceiling, so the only lever left to the experimenters is the qubit count. Random dense graphs reach the same widths as the brickwork at the same search budget (within one), so for this method the construction is already as hard as a random graph.

Two further facts follow from the stabilizer structure. The output distribution of the instance is **exactly uniform on every set of at most 14 qubits** (the minimum stabilizer weight of the graph state is exactly 15, proven by exhaustive enumeration), so no statistic on fewer than 15 qubits can see the signal, and verification of such an experiment is global by nature. Conversely any low-order bias in the hardware shots is pure noise, detectable without simulation: all 70 single-bit, 2,415 pair, 54,740 triple and 916,895 quadruple parities of IBM's post-selected shots are at chance (χ^2 p = 0.78, 0.36, 0.28, 0.95), so whatever brings the fidelity to 0.46 leaves no trace below weight 5.

The method is two standard ingredients joined by one identity: the rank-width simulability of graph-state measurements (Van den Nest et al. 2007) and perfect sampling in the style of Ferris and Vidal (2012), joined by the observation that the post-measurement states Z^v|G⟩ are orthonormal, so every prefix marginal is a plain sum of squares, Pr(b1…b_t) = 2^{-t} Σ|P_t|². We verify the sampler against dense statevectors on small instances, against an independent amplitude algorithm on the full instance, against a closed-form stabilizer-group formula for its per-step conditionals (agreement $3{\cdot}10^{-8}$ on all 1,726 shots), and across three different dynamic-program paths.

Results at a glance

	Result	Artifact
Exact samples of IBM's 70-qubit instance	**1,012 samples, 13.0 s each**, 16 cores, peak 34.4 GB	`results/samples_nq70.txt`, `time_nq70.txt`
Same instance on a laptop	**3 samples, ~18 s each**, one 8.6 GB table	`results/laptop_samples_w30.txt`
Independent engine (different algorithm)	6 samples reproduced to 10^{-6} in $\log_2$ Pr; 10-bit-flip control moves by 3.37	`results/xcheck70_independent.txt`, `xcheck70d.txt`, `xcheck70e_laptop.txt`
IBM's 1,726 hardware shots scored exactly	**linear XEB 0.466 ± 0.031**; uniform control 0.16 ± 0.12; bit-reversed control −0.15 ± 0.10	`results/ibmxeb/analysis.txt`
Agreement with the independent code of #260	1,726 of 1,726 shots within $9.4{\cdot}10^{-6}$ in $\log_2$ Pr (median $2.8{\cdot}10^{-7}$)	`results/ibmxeb/analysis.txt`
Fidelity of IBM's run from exact probabilities	0.457 ± 0.037 (XEB) and 0.48 [0.43, 0.54] (mixture MLE); IBM's DFE 0.37, ≥ 0.092	`results/ibmxeb/analysis.txt`
76-qubit, depth-76 instance sampled exactly	**19 samples, 190 s each**, 69 GB, two reproduced to 10^{-7} by the independent engine	`results/nq76/analysis.txt`
Fine-structure check without a dynamic program	stabilizer-group formula reproduces the sampler's conditionals to $3.3{\cdot}10^{-8}$ on all shots	`results/chain/stabprefix_ibm_fwd.txt`
Ceiling for every graph	width ≤ ⌊n/2⌋: 275 GB at n = 70, 281 TB at n = 90; random graphs G(70, ½) reach width 32, brickwork 31	Proposition in §5, `results/ceiling_tightness.jsonl`
Low-order marginals of the	exactly uniform on every set of ≤ 14 qubits (minimum	`results/minweight_exact.txt`, `results/minweight_k11.txt`,

ideal distribution	stabilizer weight 15, proven)	`results/minweight_case1_k11.txt`
Hardware shots, simulation-free test	70 single-bit, 2,415 pair, 54,740 triple and 916,895 quadruple parities at chance (χ^2 p = 0.78, 0.36, 0.28, 0.95)	`results/hw_marginals.txt`

1. The task

Circuit `nq70_depth70_checks27_basis_fstate`, Quantum Advantage Tracker issue #151:

1. Hadamards on 70 qubits in a line.
2. Brickwork of nearest-neighbour CZ gates to CZ depth 70, with a random √X or S√X on every qubit between layers.
3. On every qubit, T†·H·Rz(θ)·H with θ = arccos(1/√3), a rotation into the "face-state" basis that maximises stabilizer extent.
4. Measurement in Z.

The hardware run (IBM Boston, Heron R3, with 27 ancillas for spacetime Pauli checks) reports a post-selected fidelity of 0.37 from direct fidelity estimation of the underlying graph state, with a 95% lower bound of 0.092. 1,726 post-selected shots are published with the instance. The construction was chosen so that stabilizer-extent methods cost 10^7 s per sample and matrix-product-state methods 10^{25} s, according to the instance README.

2. Method

Graph form. The Clifford part maps to a stabilizer tableau, and then, by Gaussian elimination and local Cliffords, to $|S\rangle = \bigotimes U_j^\dagger |G_A\rangle$. Then

$\langle b|\psi\rangle = 2^{-n/2} \Sigma_x \Pi_j M_j[b_j, x_j] (-1)^{\Sigma_{i<j} A_{ij} x_i x_j}$, where $M_j = W_j U_j^\dagger$.

Cut-rank dynamic program. Fix a vertex order. After the prefix P is processed, the future depends only on $s = A[F,P]\cdot x_P \in GF(2)^{|F|}$. That vector lives in a space of dimension r = rank A[F,P], so the prefix is a table P[c] of 2^r complex numbers in a chosen basis of that space. Each step is a surjective GF(2)-linear map $(c, x_v) \mapsto c'$. We evaluate it in gather form: a `pdep` scatter plus a kernel of dimension at most 2. A generic matrix product state of the same cut needs bond dimension 2^r and therefore cost polynomial in 2^r; here the cost is linear in 2^r.

Exact sampling. With prefix P measured with outcomes b_P, the post-measurement state is

$2^{-|P|/2} \Sigma_c P[c] Z^{B c} |G_F\rangle$.

For $v \neq 0$, Z^v has no X part, so it anticommutes with some graph stabilizer and $\langle G|Z^v|G\rangle = 0$. The states $\{Z^v|G_F\rangle\}$ are therefore orthonormal, and $Pr(b_P) = 2^{-|P|} \Sigma_c |P[c]|^2$. Measuring qubit by qubit in the DP order:

1. compute both candidate extensions' norms (one read-only pass);
2. draw the outcome;
3. store the chosen extension (a second pass).

This is the graph-state analogue of perfect MPS sampling (Ferris & Vidal 2012). The output is the exact Born distribution, up to floating-point error. Unlike rejection sampling, the cost does not depend on the

shape of the output distribution, and unlike capped rejection sampling (#260) the output law is the Born law.

In-place layout. A second program keeps a single table and updates it in place, so the memory is 2^{width} complex64 entries rather than two tables of 2^{width+1}: 8.6 GB for the width-30 order of this instance, which is why a laptop can run it.

Cost for this instance. A greedy-plus-local-search order (`order2.py`) reaches maximum cut rank 30, the value #260 also reports; the natural order gives 32. The production sampling run used an earlier width-31 order and cost $4\cdot\Sigma_k 2^{r_k} = 6.0\cdot10^{10}$ table updates per sample with two complex64 tables of 2^{31} entries; the later runs (laptop, hardware-shot scoring, chain mode) use the width-30 order.

3. Results

3.1 Sampling IBM's instance

1,012 samples in 13,202 s on an AWS r7a.4xlarge (16 cores, AMD EPYC 9R14), peak RSS 34.4 GB, about $4.50 of on-demand time.

- All samples are distinct.
- Linear XEB against their own exact probabilities is 1.020 ± 0.047, i.e. $2^{70}\,\Sigma\,p^2 \approx 2.02$, consistent with Porter–Thomas-like output.

Laptop. With the in-place program, the arena is a single complex64 table of 2^30 entries (8.6 GB) for a width-30 order. On a laptop (AMD Ryzen 9 9955HX, 16 threads, container memory capped at 10 GB), 3 exact samples took 55 s including allocation, i.e. **about 18 s per exact sample** (`results/laptop_samples_w30.txt`). The in-place sampler was validated on every bitstring of small circuits. On the full instance it reproduces the cloud samples' $\log_2$ Pr to $3\cdot10^{-7}$, and the independent amplitude engine confirms the first laptop sample: −68.5019327 against −68.5019326 reported (`results/xcheck70e_laptop.txt`).

3.2 IBM's hardware shots, scored exactly

The sampler's probability mode evaluates the exact Born probability of a given bitstring through the same code path as sampling. We scored all 1,726 hardware shots published with #151 on an r7a.4xlarge: 6.9 s per shot, 3 h 19 min in all, 17.2 GB peak (`results/ibmxeb/`). The run first reproduced two earlier cloud samples' $\log_2$ Pr to $5\cdot10^{-7}$ as a gate.

strings	N	linear XEB
IBM hardware shots	1,726	**+0.466 ± 0.031** (bootstrap 95% [+0.41, +0.53])
uniform random strings (control)	64	+0.16 ± 0.12
IBM shots with the bit order reversed (control)	64	−0.15 ± 0.10

Our $\log_2$ Pr agrees with #260's independently written FP64 code on every one of the 1,726 strings, to $9.4\cdot10^{-6}$ at worst (median $2.8\cdot10^{-7}$), and the XEB of +0.4660 is the same to four digits. Two programs written independently assign the same probability to each of IBM's shots.

Normalising by the ideal value $2^{70}\Sigma p^2 - 1 = 1.020 \pm 0.047$, measured on our own exact samples, gives a fidelity estimate F_XEB = 0.457 ± 0.037. A second estimator uses the whole distribution of $z = 2^{70}\cdot p(x)$ over the shots rather than its mean. If each shot is drawn from the ideal distribution with probability F and uniformly otherwise, the density of z over shots is $(Fz + 1 - F)e^{-z}$, and the maximum-likelihood fit gives F_MLE = 0.48 [0.43, 0.54] (profile likelihood, 95%). The mixture fits the observed shot

distribution (Kolmogorov–Smirnov distance 0.018, 5% threshold 0.033; `ibm_fidelity_mle.py`). On synthetic shots of known fidelity the two estimators return 0.00 ± 0.02 and 0.00 [0.00, 0.05] for F = 0, 0.42 ± 0.04 and 0.41 [0.35, 0.46] for F = 0.4, and 0.94 ± 0.05 and 0.98 [0.95, 1.00] for F = 1 (`fidelity_mle_synth.py`, `results/fidelity_mle_synthetic.txt`).

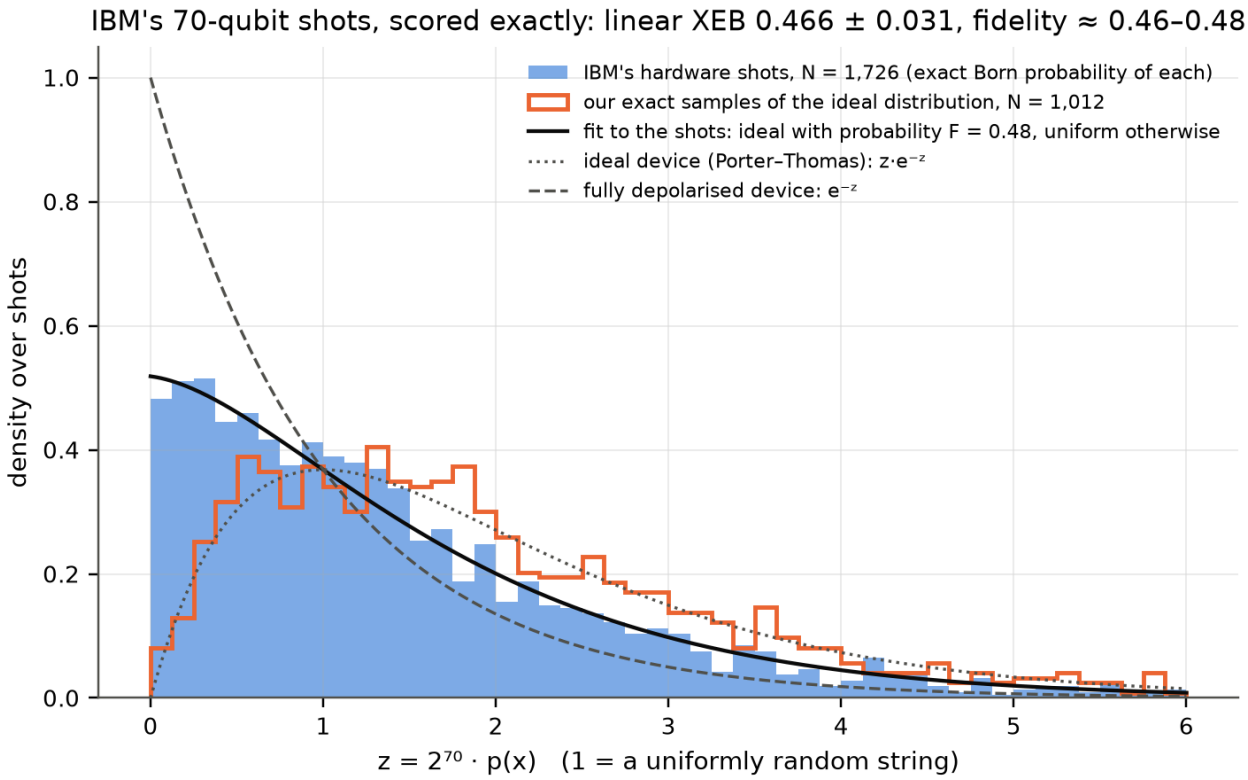


Figure 2. Distribution of $z = 2^{70} \cdot p(x)$ over IBM's 1,726 post-selected shots (exact p), our 1,012 exact samples, and the fitted clean/uniform mixture (`paper/make_figures.py`).

The instance README (tracker repository, `random_graph_sampling/README.md`) states that "XEB requires classical simulations for the outcome probabilities, hence can only be extrapolated for the full-depth circuit". Here it is computed exactly for every published shot. The shots are post-selected by spacetime Pauli checks on 27 ancillas, and the README reports a post-selected state fidelity of 0.37 by direct fidelity estimation with 80 random stabilizers, "bounded above 0.092 with 95% confidence". Our two estimates, 0.457 ± 0.037 and 0.48 [0.43, 0.54], are consistent with that bound and higher than the point estimate, with a much narrower interval. The quantities differ (the full circuit including the non-Clifford measurement layer, versus the stabilizers of the graph state; the README argues they agree to first order under Pauli twirling), and linear XEB is known to over-estimate fidelity under non-depolarising noise, so we report the gap without interpreting it. It is, in our view, the most useful thing in this note for the experimenters: an exact, per-shot benchmark of their device at full depth.

3.3 A 76-qubit instance, sampled

To test the frontier with a measurement rather than a projection, we generated a 76-qubit, CZ-depth-76 instance of the same construction (`frontier_instance.py`, seed 1076, 1,465 edges), found an order of width 33 (natural order 38), and ran the in-place sampler on the same 16-core node class (r7a.4xlarge, 128 GiB): **19 exact samples in 3,604 s, 190 s per sample, 69 GB peak** (`results/nq76/`). The node first reproduced the two 70-qubit reference probabilities as a gate. All 19 samples are distinct; their self-consistency XEB is 0.83 ± 0.22. The cross-check scheduled in that run produced nothing (its complex128 tables, 137 GB, were killed by the kernel; `results/nq76/run.log`), so it was repeated in a second run. The independent sliced amplitude engine, rebuilt with complex64 tables and itself gated on

the two known 70-qubit values (−69.5085289 and −68.8761655 against −69.5085302 and −68.8761651), recomputes the first two samples' $\log_2 \Pr$ as −74.1389442 and −75.3402128 against the sampler's −74.1389441 and −75.3402127 (`results/nq76/analysis.txt`). An 80-qubit instance (seed 1080) has an order of width 35, a 275 GB table; its inputs are prepared and the run waits on cloud capacity.

4. Verification

1. **Small instances.** On circuits of the same form with up to 9 qubits, the distribution implied by the sampler matches the statevector on every bitstring to $2{\cdot}10^{-15}$. The C++ sampler's 20,000-shot histogram has the same total-variation distance from the exact distribution as an ideal sampler's. Every test has a negative control that fails as intended.
2. **Full instance.** A sliced amplitude engine recomputes $\log_2 \Pr$ of sampled bitstrings; it is a different algorithm from the sampler.

sample	sampler $\log_2 \Pr$	independent engine $\log_2 \Pr$
1	−69.5085302	−69.5085294
2	−68.8761651	−68.8761651
3	−68.3318209	−68.3318207
4	−69.0978566	−69.0978571
5	−70.2946875	−70.2946876

Control: sample 1 with 10 bits flipped gives −72.8808316. That is 3.37 lower in $\log_2$ (10× less likely), against agreement within 1e-6 on the real samples.

1. **Reproducibility.** The production run first reproduced, bit for bit, three samples of a 55-qubit sub-instance (width 26) generated on a different machine, compiler and thread count (`results/check55_result.txt`).
2. **Engine-independent check.** While the cut rank of the measured prefix equals its length, the reduced state is maximally mixed, so the prefix marginal is exactly uniform; for the production order that holds for the first 30 steps. Over the 1,012 samples the first 30 step-bits show pair parities (435 tests) and triple parities (4,060 tests) at chance (4.8% and 4.4% beyond 2σ, 4.6% expected; largest $|z|$ 3.2 and 3.8). Of the 30 single-bit means, 4 fall beyond 2σ ($p \approx 0.05$), the largest at 2.4σ. A control in which one bit is replaced by a copy of another fails the test (`check_uniform_prefix.py`, `results/uniform_prefix_check.txt`). This uses only the rank profile and the sample file.
3. **Stabilizer-only conditionals.** Once the cut rank falls below the prefix length, the prefix marginal has a closed form from the stabilizer group alone: $\Pr(b_F) = 2^{-t} \Sigma_S \mathrm{sign}(S) \Pi_{j\in \mathrm{supp}\, S} (-1)^{b_j} n_j[S_j]$, summed over the $2^{t-\mathrm{rank}}$ stabilizers supported on F, with n_j the Bloch vector of the measured operator. We enumerate that sum exactly while the deficit is at most 16 (steps 1–42 of the width-30 order) and compare the resulting conditionals $\Pr(b_k \mid b_1 \ldots b_{k-1})$ with the sampler's chain output for all 1,726 hardware shots: the two agree to $3.3{\cdot}10^{-8}$ (half a float32 ulp at ½, the sampler's storage precision), while the conditionals themselves depart from ½ by up to $8{\cdot}10^{-5}$ at those steps (`stabprefix.py`, `results/chain/stabprefix_ibm_fwd.txt`). On small circuits the same formula reproduces the statevector marginals to $6{\cdot}10^{-15}$ on 162 prefixes, and a flipped generator sign fails it. This is a check of the dynamic program's fine structure by a method that never builds the DP table.

The same formula explains where the information in a string lives: a stabilizer of weight w contributes at most $\max|n|^w$, so the conditionals stay within 10^{-4} of ½ until the prefix supports many low-weight stabilizers, which in this order happens only in the last ~10 steps; the per-step spread of the chain conditionals grows by about 1.4× per step from $7{\cdot}10^{-4}$ at step 50 to 0.56 at step 70, and the last 8 steps carry 2.83 of the 2.85 variance of $\log_2 \Pr$ across the hardware shots (`results/chain/analysis_profile.txt`). For our 1,012 exact samples the picture is the same: the spread

of the conditionals first exceeds 10^{-3} at step 51 (52 in the reversed order), and the last 4 and 8 steps account for 90% and 100% of the variance of $\log_2 \Pr$ (`results/chain/analysis_profile.txt`).

1. **Three dynamic-program paths agree.** Scoring every string along the width-30 order, along its reverse (the same width, a different table layout and step sequence), and, for our samples, along the width-31 production order gives the same $\log_2 \Pr$: the forward and reversed orders agree to $1.4 \cdot 10^{-5}$ at worst (median $4 \cdot 10^{-7}$) on the 1,726 hardware shots and to $3.5 \cdot 10^{-6}$ on the 1,012 samples, and the width-31 run agrees with the width-30 order to $4.3 \cdot 10^{-6}$ (`results/chain/analysis_crossorder.txt`). The stabilizer-only check of item 5 also holds along the reversed order and on the ideal samples ($3.3 \cdot 10^{-8}$, `results/chain/stabprefix_rev_ideal.txt`).
2. **Simulation-free test, and the sampler's random stream.** By §6 every marginal of the ideal distribution on at most 14 qubits is exactly uniform, so all single-bit means, pair parities and triple parities are exactly zero (for ±1 bits), and their sums of squared z-scores are χ^2 with as many degrees of freedom as tests. Our 1,012 samples: pairs $p = 0.76$, triples $p = 0.97$, quadruples $p = 0.12$, single bits $\chi^2 = 100$ on 70 ($p = 0.010$); the same run scores truly uniform random bits as a calibration and a control with one bit biased by 15% fails it ($p = 1.8 \cdot 10^{-7}$) (`hw_marginals.py`, `results/hw_marginals.txt`). We examined the single-bit excess. For the first 30 steps of the production (width-31) order the cut rank equals the prefix length, so the conditional is exactly ½ and the sampled bit must equal ($U < ½$) for the sampler's uniform draw U; reproducing the stream (std::mt19937_64 seeded with $10^6 + i$, `rngbits.cpp`) gives exactly that bit in 1,012 of 1,012 samples, and in 0 of 1,012 with the seed shifted by one (`results/rng_check.txt`). The sampler is therefore deterministic given its stream. Over those 30 exact steps the single-bit χ^2 is 36.9 on 30 ($p = 0.18$); the excess sits in steps 31–70 (63.5 on 40, $p = 0.010$). The sampled bit still coincides with the stream's sign bit through step 58 of the production order (`results/rng_check.txt`), as expected if its conditionals, like those of the width-30 order profiled in Figure 3, stay within 10^{-3} of ½ until about step 50; so a χ^2 of the stream alone over the same seeds is not independent evidence. What is independent is a direct test of the drawing step with the exact conditionals: the residuals $b_k - \Pr(b_k = 1 \mid \text{prefix})$ have mean zero and variance $p(1 - p)$ whatever the prefix, so their per-step z-scores are standard normal if each bit is drawn with its own conditional (`draw_check.py`, `results/draw_check.txt`, on the width-30 chain output of item 6). Over the 20 steps where the conditionals carry the information (51–70) this gives $\chi^2 = 23.1$ on 20 ($p = 0.29$); over steps 23–50, where the conditionals differ from ½ by less than 10^{-3}, 44.2 on 28 ($p = 0.027$); over all 70 steps 94.7 ($p = 0.026$). The test is wired to the conditionals: with the conditionals flipped ($p \to 1 - p$) on the real samples it fails at $p = 2 \cdot 10^{-4}$, a synthetic sampler drawing with p^{1.1} instead of p fails at $p = 7 \cdot 10^{-50}$, and correct synthetic draws from the same conditionals pass ($p = 0.92$). A scan of 300 blocks of 1,012 consecutive seeds shows no bias in the generator (mean χ^2 over 70 draws 70.3 ± 0.7; `results/rng_scan.txt`). The excess therefore sits in bits that are, to within 10^{-3}, the generator's own sign bits, and the drawing step passes where the dynamic program's conditionals matter. We record the single-bit $p = 0.010$ as a flag, not an explanation.

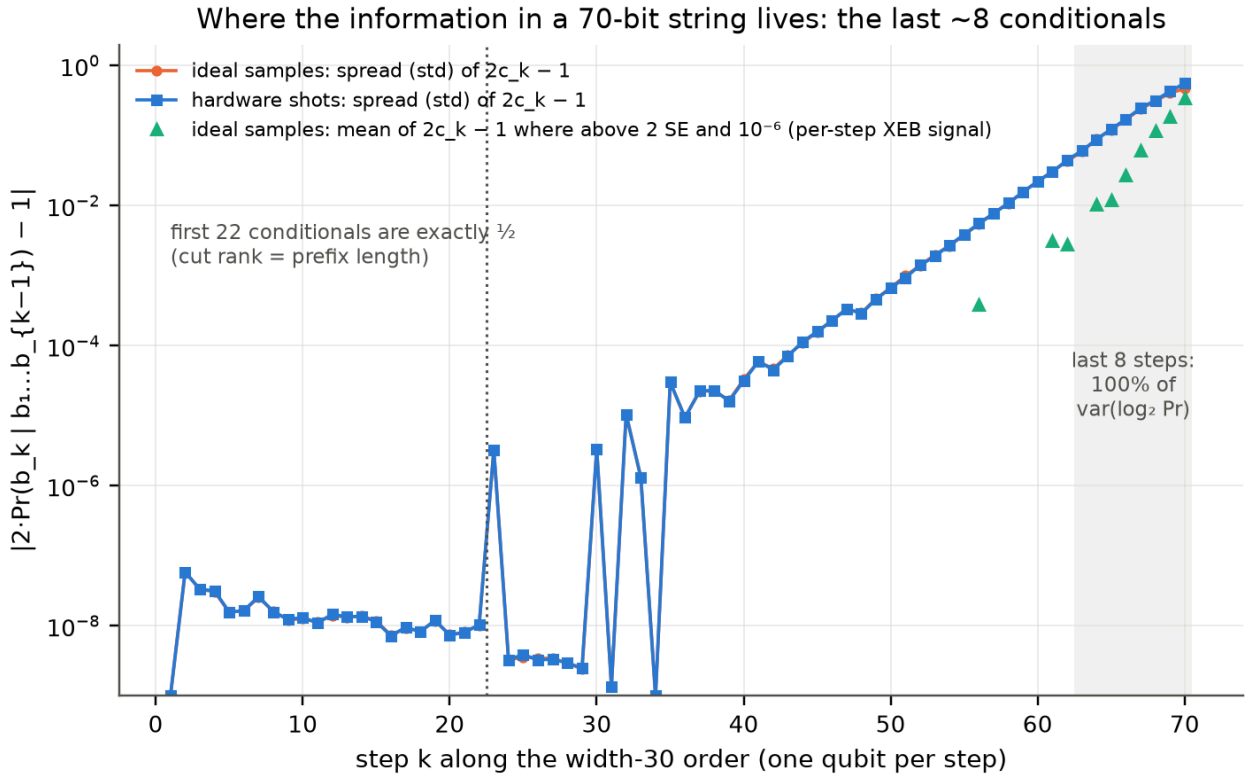


Figure 3. Spread of the exact per-step conditionals along the width-30 order for the hardware shots and the ideal samples, and the mean of 2c_k − 1 over ideal samples where it is resolved (the per-step XEB signal). The first 22 conditionals are exactly ½; the information sits in the last ~8 steps.

Per-step fidelity of the hardware shots. Dividing the hardware mean of 2c_k − 1 by the ideal mean gives a fidelity-like ratio for each informative step. Because the information sits in the last steps, only four steps per order are informative (ideal mean above 0.05): in the forward order, qubits 9, 8, 13 and 50 give 0.45 ± 0.12, 0.54 ± 0.08, 0.52 ± 0.07 and 0.48 ± 0.04; in the reversed order, qubits 45, 68, 53 and 21 give 0.47 ± 0.10, 0.43 ± 0.07, 0.43 ± 0.06 and 0.53 ± 0.05. All eight are consistent with the global fidelity estimates of 0.46–0.48 and none stands out, so the shots show no sign of a qubit that is much worse than the rest among those eight. A per-qubit map of all 70 qubits would need an order ending in each group of about four qubits, roughly 18 further width-30 passes; we did not run them.

5. Hardness frontier

We generated circuits with the #151 construction (linear chain, alternating CZ brickwork with random √X or S√X layers, CZ depth n) for n = 30–120, with one instance per n, and ran the same order search (`frontier.py`, output `results/frontier.jsonl`). At n = 70 the generated instance matches the real one: 1,192 vs 1,150 edges, natural-order width 33 vs 32, best width 31 vs 31 at the same search budget (a longer search on the real instance reaches 30).

n	best width found	memory (2 complex64 tables)	table updates per exact sample
30	12	< 1 MB	2^16.7
50	22	67 MB	2^26.4
60	26	1.1 GB	2^30.8
70	**31**	**34 GB**	**2^36.0**

76 (sampled)	33	69 GB in-place	2^38.2
80	36 (35 found later)	1.1 TB (275 GB in-place)	2^41.2
90	42	70 TB	2^46.3
100	47	2.3 PB	2^51.4
110	51	36 PB	2^55.9
120	56	1.2 EB	2^61.2

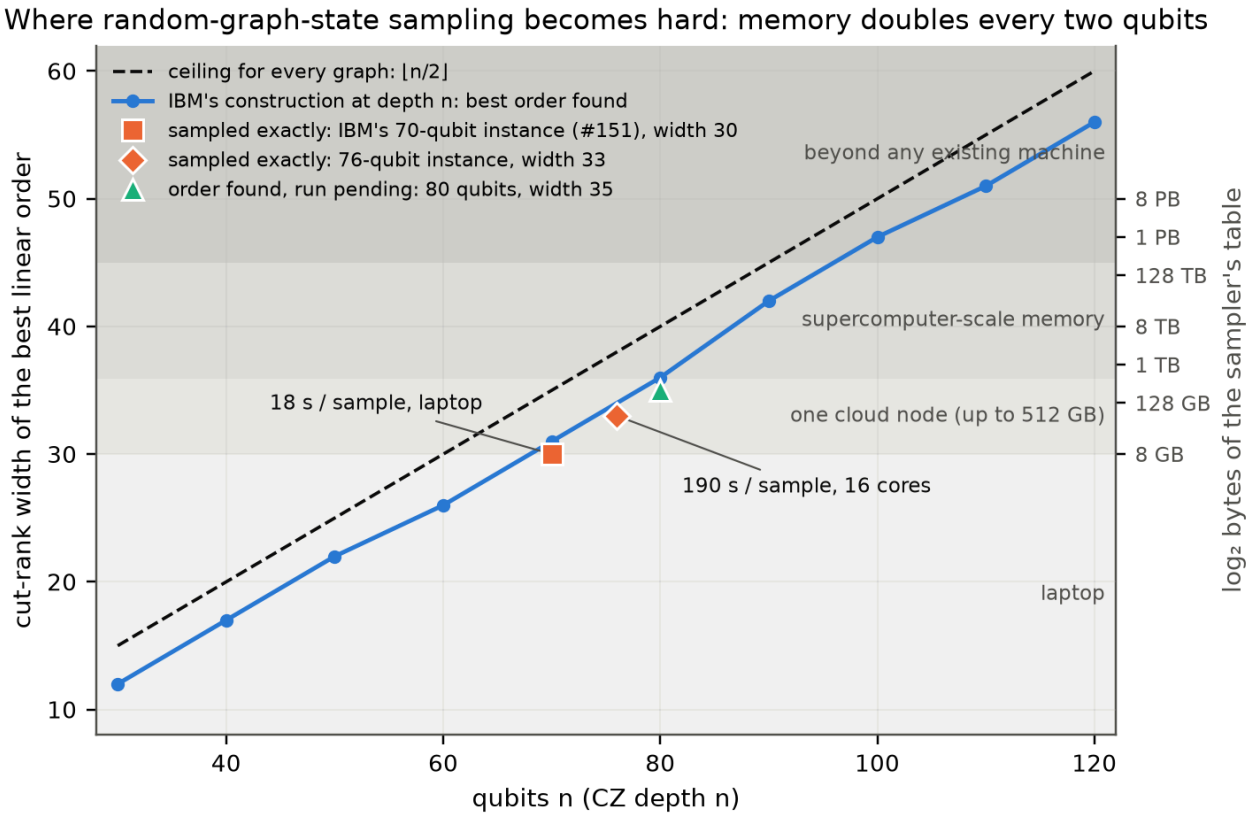


Figure 1. Best width found versus n for the construction, the ⌊n/2⌋ ceiling that holds for every graph, the two instances sampled exactly, and the memory of the in-place table. Shaded bands mark what a laptop, one cloud node, a supercomputer, and no existing machine can hold. The 76- and 80-qubit order searches are in `results/frontier_instances/` *(fi76_s2: width 33, log₂ Σ_k 2^{r_k} = 36.2, i.e. 2^{38.2} updates; fi80_s5: width 35).*

The width fits 0.492·n − 2.9; at every n it is between n/2 − 4 and n/2 − 3. So memory doubles with every two qubits. At depth n, this method moves from one server (n = 70) to one large-memory node (n ≈ 80), to a leadership-class supercomputer (n ≈ 90), and past any existing machine (n ≳ 100).

Proposition (a ceiling for every graph). *Let |G⟩ be a graph state on n qubits and let each qubit be measured in an arbitrary single-qubit basis. For every linear order of the qubits, the cut rank of every prefix is at most ⌊n/2⌋. Consequently the sampler above draws an exact Born sample with at most 2^{⌊n/2⌋} complex table entries and at most 4·n·2^{⌊n/2⌋} table updates, whatever the graph and whatever the measurement bases.*

Proof. The cut rank of a prefix P is the GF(2) rank of the |F| × |P| adjacency block A[F,P], which is at most min(|P|, |F|) = min(t, n − t) ≤ ⌊n/2⌋. The in-place sampler's table has 2^{r_k} entries at step k and each step makes at most four passes over it. ■

The proposition is elementary, but its consequence is not widely stated: at n = 70 the ceiling is 275 GB and about $9.6 \cdot 10^{12}$ updates per sample; at the rates measured in this work ($6.0 \cdot 10^{10}$ updates in 13.0 s for the two-table sampler, 2^{38.2} updates in 190 s for the in-place sampler at 76 qubits, i.e. $4.6 \cdot 10^{9}$ and $1.7 \cdot 10^{9}$ per second on 16 cores) that is between 35 minutes and 1.6 hours per sample; at n = 76 it is 2.2 TB (the largest single cloud nodes); at n = 90 it is 281 TB (the aggregate memory of a leadership machine); at n = 100 it is 9 PB. So no graph-state sampling experiment with product measurements can escape a single large-memory node below about 76 qubits, or a supercomputer below about 90, no matter how the graph is chosen. The random brickwork construction already sits within 3–4 of this ceiling (Figure 1), so better graphs cannot buy the experimenters more than a factor of about 10 in memory; the only lever left is the qubit count.

Is the ceiling tight? Running the same order search (four randomised greedy orders then 3,000 local-search moves, two instances per cell; the command is recorded in `results/ceiling_tightness.cmd.txt`) on Erdős–Rényi graphs G(n, ½) and G(n, 0.3) and on the brickwork construction gives (`ceiling_tightness.py`, `results/ceiling_tightness.jsonl`):

n	ceiling ⌊n/2⌋	G(n, ½)	G(n, 0.3)	brickwork
40	20	16, 16	16, 16	16, 17
50	25	21, 21	22, 21	22, 21
60	30	26, 26	26, 26	26, 26
70	35	32, 32	32, 31	31, 31

Random graphs of either density sit 3–4 below the ceiling and the brickwork within one of them, so for this method the construction is already as hard as a random graph, and the ceiling is tight to within the heuristic's few bits.

Caveats. The widths are upper bounds from heuristic orders; better orders or tree decompositions would lower them. Slicing trades memory for time at a measured cost of about 1.5× per sliced vertex on this instance, not 2×. The table has one instance per n. The ceiling bounds this family of methods, not all classical methods, and says nothing about circuits that are not graph states followed by product measurements: the doped sibling instance on the tracker, with 468 T gates spread through the circuit, is outside its reach (its amplitudes were computed by tensor-network methods in arXiv:2608.13110).

6. Where the signal is: no marginal on fewer than 15 qubits carries any

In the graph frame every stabilizer is S_v = X^v Z^{Av} with support v ∪ Av, and by the formula of §4 item 5 the marginal of the output distribution on a qubit set F is Pr(b_F) = 2^{-|F|} Σ_{S supported on F} sign(S) Π_j (−1)^{b_j} n_j[S_j]. The measured Bloch vectors have no zero component (all three components are ±1/√3 in the graph frame), so:

Proposition. *For the face-state basis (every Bloch component ±1/√3), the marginal on F is exactly uniform if and only if no nontrivial stabilizer is supported inside F. In particular every marginal on fewer than d_min qubits is exactly uniform, where d_min is the minimum weight of a stabilizer of the graph state, and a stabilizer of weight w changes a marginal by at most the fraction 3^{−w/2} of its uniform value.*

Proof. "If" is the formula with the identity as the only term. For "only if", let U be a minimal nonempty support among the stabilizers inside F and H_U the subgroup of stabilizers with support inside U; by minimality every nontrivial element of H_U has support exactly U. For any j ∈ U the map S ↦ S_j (the Pauli letter at j, modulo phase) is a homomorphism from H_U to the one-qubit Pauli group modulo phases whose kernel consists of elements with identity at j, hence of support smaller than U, hence

trivial; so H_U embeds in the Klein group and has 1 or 3 nontrivial elements. The Fourier coefficient of the marginal at the character Π_{j∈U}(−1)^{b_j} is the sum over exactly those elements of sign(S) Π_j n_j[S_j] = ±3^{−|U|/2}, an odd number of terms of equal magnitude, which cannot vanish. ■ (For a general product basis with nonzero Bloch components "if" still holds and "only if" holds unless three such terms cancel exactly.)

For IBM's instance **d_min = 15 exactly**, by a three-way split that is exhaustive (`minweight_exact.py`, `results/minweight_exact.txt`): a stabilizer of weight ≤ 14 has either at most 11 X-support vertices (all $2.6{\cdot}10^{12}$ such v enumerated, lightest 15; `minweight.cpp`, `results/minweight_k11.txt`), or at most 11 Z-support vertices c = Av (all such c enumerated through a generalised inverse of the rank-68 adjacency and its 4-element kernel coset, lightest 15; `minweight_case1.cpp`, `results/minweight_case1_k11.txt`), or at least 12 of each, in which case v and Av differ in at most 4 positions and v is recovered from (A+I)v = e for the 974,120 vectors e with |e| ≤ 4 (lightest 16; A+I has full rank, so e = 0 contributes nothing and each e has one solution). The first enumeration alone cannot certify the minimum, which is why its output file ends "NOT PROVEN"; the certificate is the union of the three. The lightest stabilizer has v = {0, 5, 27, 42, 48, 49, 52, 58, 61} and support of 15 qubits. The split was validated on the 30- and 40-qubit generated instances, whose minima (6 and 8) are known from direct enumeration: it returns the same minima, and on the 30-qubit instance, run with t = 2 so that the minimiser lies outside the directly enumerated part, only case (2) finds the weight-6 stabilizer, so dropping that case would report 8 (at 40 qubits, t = 4, the direct part finds the minimiser and the other two cases return 9) (`minweight_split_test.py`, `results/minweight_split_test.txt`). So **every marginal on at most 14 qubits is exactly uniform**, and even the first non-uniform marginals deviate by at most about $3^{-7.5} \approx 3{\cdot}10^{-4}$ per stabilizer. This is the stabilizer-side statement of Figure 3: no test restricted to 14 or fewer qubits, however many shots, can tell the ideal output from coin flips; the signal lives in correlations of weight ≥ 15, and verifying such an experiment is global by nature.

The threshold grows with the size of the experiment. For generated instances of the same construction (`frontier.py` seeds, exhaustive enumeration with the proof condition met at every size; `minweight_scan.py`, `results/minweight_scan.txt`):

n	30	40	50	60	70 (IBM's instance)
minimum stabilizer weight d_min	6	8	9	12	15

So the smallest marginal that carries any signal grows roughly like n/5: at 100 qubits one would expect every marginal on about 20 qubits to be exactly uniform.

The same fact turns the hardware shots into a simulation-free diagnostic. Any deviation of a ≤ 14-qubit marginal of the device from uniform is noise, measurable from the shots alone. For IBM's 1,726 post-selected shots all 70 single-bit means (χ^2 61 on 70, p = 0.78), 2,415 pair parities (p = 0.36, largest |z| 4.4), 54,740 triple parities (p = 0.28) and 916,895 quadruple parities (p = 0.95, largest |z| 5.6, which is p ≈ 0.02 after correcting for the number of tests) are at chance (`results/hw_marginals.txt`). The errors that bring the fidelity to about 0.46 therefore leave no low-order trace at all: no biased qubit, no correlated pair, no correlated triple or quadruple; after the Pauli checks the residual noise acts on the output like a global depolarisation (Figure 4).

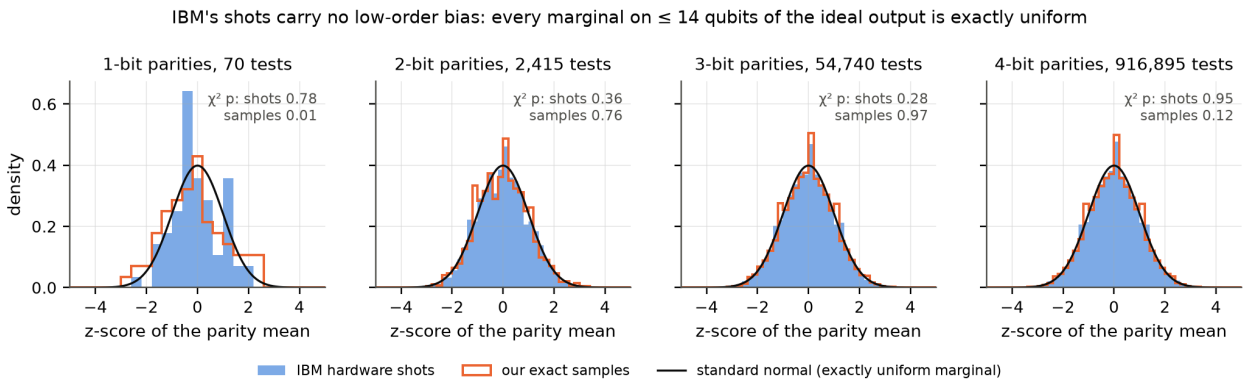


Figure 4. Distribution of z-scores of all 1-, 2-, 3- and 4-bit parity means over IBM's 1,726 shots and over our 1,012 exact samples, against the standard normal that an exactly uniform marginal implies. The χ^2 p-values are those quoted in this section for the shots and in §4 item 7 for the samples (`hw_marginals.py`). This is consistent with the mixture fit of §3.2 and is, we think, a useful fact for the experimenters.

7. The doped instance

IBM's flagship variant, `nq70_depth70_checks27_doped`, inserts 468 T gates throughout the same 70-qubit brickwork and measures in Z; its README calls it adversarial for both tensor-network and stabilizer methods (best known stabilizer rank 2^{185.5}) and reports a hardware fidelity of 0.26 ± 0.02 after post-selection. The only classical entry on the tracker computes exact amplitudes by tensor-network contraction (#242: contraction width 35, 2^{48.1} complex FMAs per batch of 256 amplitudes, all 2,051 IBM shots in 37.3 minutes on 32 nodes of eight H100 GPUs, log-XEB 0.350 ± 0.027); nothing has sampled it.

T gates as projected ancillas. For a data qubit q, a fresh ancilla a in |0⟩, CNOT(q → a), and the projection of a onto |φ⟩ = (|0⟩ + e^{−iπ/4}|1⟩)/√2 give ⟨φ|_a CNOT |ψ⟩_q|0⟩_a = T|ψ⟩/√2 for every |ψ⟩ (the orthogonal outcome gives ZT|ψ⟩/√2, so each projection succeeds with probability exactly ½ whatever the state). The projection is a product measurement: apply T then H to the ancilla and keep outcome 0. Hence the doped circuit is a Clifford circuit on 70 + 468 = 538 qubits followed by a product measurement, a graph state measured in a product basis, and Pr(all 468 projections succeed) = 2^{−468} exactly, so Pr_doped(b) = 2^{468}·Pr_gadget(b, anc = 0). We checked the identity on 20 random circuits against dense statevectors (1e-10), the whole pipeline on 25 more (1e-15), and the sampler binary itself on 12 (1e-8), each with a control that drops the ancilla's T (`doped_gadget.py`, `test_doped.py`, `test_gsamp2_doped.py`).

Width. On the 538-vertex graph the natural order has width 137, but a spatial sweep (qubit 0 to 69, each qubit's ancillas in time order then its data vertex) has width **34** (`results/doped/doped_widths.json`): the cut between the left and right halves of the chain is crossed only by the CZ gates on one bond, at most 35 of them, and the sweep realises nearly that bound; simulated annealing from the sweep and single- and double-vertex slice searches found nothing lower. One exact amplitude therefore costs a table of 2^{34} complex64 (128 GiB) and 2^{40.9} table updates. For comparison, #242's contraction has width 35 and costs 2^{48.1} complex multiply-adds per batch of 256 amplitudes, 2^{40.1} per amplitude: the two methods do the same amount of arithmetic per amplitude to within a factor of two, and the gap in wall-clock time (their 37 minutes for all 2,051 shots on 256 H100 GPUs against our 82 minutes per shot on one 32-core node) is hardware, not algorithm.

Why the chain rule fails here, and what replaces it. The orthonormality identity traces out the unmeasured suffix, but the ancillas must be projected, not traced (a traced ancilla dephases its T gate).

Putting all ancillas before any sampled data qubit makes the prefix–suffix cut contain the whole ancilla|data cut, whose rank is 70; exact chain-rule sampling would need 2^{70} entries. Instead we move the last m = 20 data vertices to the end of the sweep order, which leaves the width at 34 and makes the final cuts 20, 20, 19, …, 1, and sample in two stages: a proposal draws the first 50 data bits uniformly; one pass gives their exact marginal Pr(prefix, anc = 0) (the 20 unmeasured data qubits are traced by orthonormality) together with exact draws of the last 20 bits from their conditionals; the proposal is accepted with probability z/c where z = 2^{50}·2^{468}·Pr(prefix, anc = 0) is the marginal in units of the uniform distribution. Because z averages 2^{20} output probabilities it is concentrated near 1 (relative spread about 2^{−10}), so with c = 1.03 almost every proposal is accepted, and the accepted output follows the Born distribution exactly except for the mass of prefixes with z > c, which is recorded for every proposal and, under the Porter–Thomas model, is below 10^{-30}. On small T-circuits the hybrid sampler reproduces the exact output distribution (total variation at the level of an ideal sampler of the same size) and a cap set below the true maximum of z is detected as a biased output (`test_hybrid.py`). At full scale we ran it on the 70-qubit fstate instance, where the exact sampler is the ground truth: 150 proposals with the width-30 plan (50 bits proposed, 20 drawn exactly), 147 accepted, z between 0.9970 and 1.0025 (spread 0.0010, as the 2^{−10} estimate predicts), no proposal above the cap; the accepted samples' XEB against their own exact probabilities is 1.018 ± 0.128 against the exact sampler's 1.020 ± 0.047, their mean $\log_2$ Pr −69.37 ± 0.09 against −69.40 ± 0.04, and three accepted samples re-scored in prob mode agree with the sampler's reported $\log_2$ Pr to $9\cdot10^{-7}$ (`results/fstate_hybrid_check.txt`, `results/fstate_hybrid_samples.txt`). Each sample costs about one width-34 pass.

Full scale: the first IBM doped shot, five ways. The first of IBM's 2,051 published doped shots was scored on three rented nodes with three different arrangements of the same computation (`graphsamp/results/doped/`, each value read from that run's output file):

run	node	plan	$\log_2$ Pr(shot 1)	seconds per pass
7	r5a.8xlarge, 256 GiB	width 34, m = 20, unsliced	−68.295330880	9,872
7	same	width 34, m = 10 (different order)	−68.295331335	—
9	r7iz.8xlarge, huge pages	m = 20 / m = 10	−68.295331925 / −68.295330954	4,919
10	r7a.4xlarge, 128 GiB	two slices at ancillas 135, 104 (4 plans of width 33)	−68.295331107	5,888 (4 slice passes)

The five values span $1.1\cdot10^{-6}$ in $\log_2$, the rounding level of complex64 tables accumulated in different orders. The sixth value comes from a different method altogether: the tensor-network entry #242 published its amplitudes (Zenodo 10.5281/zenodo.21912448; q0-first strings, which are our Qiskit strings reversed, all 2,051 matching), and its shot 1 has $\log_2$ p = −68.295074. The relative difference between the two methods is $1.8\cdot10^{-4}$, within the precision of a 2^{48} complex64 contraction (`doped_tn242_compare.py`, `results/doped/tn242_crosscheck_run7.txt`). This is, to our knowledge, the first time two independent exact methods have been compared string by string on this circuit.

What the published exact probabilities say about the hardware run. Since our values agree with #242's, their 2,051 probabilities can be analysed with the estimators of §4 (`ibm_fidelity_mle.py`, `doped_tn242_analysis.py`; outputs in `results/doped/tn242_fidelity.txt` and `tn242_porter_thomas.txt`). Each of their batches also holds the 255 sibling strings that differ from the shot on qubits 0–7, so the ideal distribution itself can be tested: over 523,005 siblings z = 2^{70}p has mean 0.9991 ± 0.0014, mean z^2 2.0013 and KS distance 0.0016 to Exp(1) (5% level 0.0019), so unlike the Clifford instance the

doped circuit's output is Porter–Thomas and E_ideal[z] = 2 is justified. For the shots: linear XEB fidelity 0.342 ± 0.028, mixture maximum likelihood 0.343 with 95% interval [0.293, 0.393], and the KS distance of the shots' z to the fitted mixture (F z + 1 − F)e^{−z} is 0.014 (5% level 0.030), so the depolarizing mixture describes the data; the log-XEB of #242 is 0.350 ± 0.027. A local check uses the conditional distribution of qubits 0–7 given the other 62 bits: the shots' mean of 256·p(x)/Σ_siblings p − 1 is 0.339 ± 0.028, the same fidelity read off eight qubits at a time. All three exceed the README's 0.26 ± 0.02, which was obtained on the device by other means, and agree with each other.

{Pending from runs 9–11: exact probabilities of further IBM shots with uniform controls (each compared with #242's value), the hybrid samples with their acceptance rate and maximum z, and the two-table engine's time per pass.}

8. What this does and does not show

It shows:

- the 70-qubit instance is sampled exactly, from the Born distribution, on a laptop, so it does not demonstrate a task beyond classical reach;
- the published hardware data can be benchmarked exactly at full depth, which the experiment's own README describes as impossible, and the resulting fidelity estimate is tighter than the one obtained on the device;
- the same construction stays within one node's reach to about 80 qubits (measured at 76), within a supercomputer's to about 90, and that no choice of graph changes this by more than a few qubits (random graphs measured within one of the brickwork);
- the output distribution is exactly uniform on every set of at most 14 qubits, and IBM's shots show no low-order bias at all.

It does not show:

- a new complexity-theoretic result; the rank-width reduction is from 2007 and the ceiling is elementary;
- priority on attacking this instance; #260 applied the same reduction with capped rejection sampling the day before our first run, and we cite it throughout;
- anything about the doped instance, or about graph-state experiments with entangled measurements.

What we would suggest the experimenters run next. At depth n with n ≥ 100, the ceiling alone puts any graph beyond rank-width methods (9 PB), with no dependence on the particular graph; alternatively, doping with a few hundred non-Clifford gates, as in the sibling instance, defeats this method outright. In either case the exact scorer here can still compute full-depth XEB for any graph-state run up to about 80 qubits, so an experiment at that scale can report an exact benchmark instead of an extrapolated one.

9. Related work

- Rank-width simulability of graph-state measurements: Van den Nest, Dür, Vidal, Briegel, PRA 75, 012337 (2007).
- Perfect sampling of matrix product states: Ferris & Vidal, PRB 85, 165146 (2012).
- ZX-calculus rank-width contraction: Kuyanov & Kissinger, arXiv:2603.06764.

- Exact amplitudes of IBM's doped circuits by T-injection and rank-width: Renaud, `dcs-rankwidth`; tracker #249.
- Capped rejection sampling for this instance on GPUs: tracker #260.
- Exact tensor-network amplitudes for the doped 70-qubit instance: Manabe, Gu, Pan, arXiv:2608.13110; amplitudes and contraction paths at Zenodo 10.5281/zenodo.21912448 (CC BY 4.0), used in §7 for the string-by-string comparison.
- The random-graph-state hardness argument behind the construction: arXiv:2412.07058.

Appendix: artifacts

All paths are relative to the repository root.

- `results/samples_nq70.txt`, `run.log`, `time_nq70.txt`, `inputs.sha256`, `check55_result.txt`: the production sampling run and its reproduction check.
- `results/xcheck70_independent.txt`, `xcheck70d.txt`, `xcheck70e_laptop.txt`: independent recomputation of five cloud samples, the 10-bit control, and one laptop sample.
- `results/laptop_samples_w30.txt`: the laptop run (3 samples, timing).
- `results/ibmxeb/`: exact log2 Pr of the 1,726 hardware shots and the controls, timing, gate, input hashes, and `analysis.txt` (XEB, comparison with #260, fidelity estimators).
- `results/chain/`: per-step conditionals of all shots and samples along two orders, the stabilizer-only comparison, the cross-order comparison and the information profile.
- `results/nq76/`: the 76-qubit samples, timing, gate, independent check and `analysis.txt`.
- `results/uniform_prefix_check.txt`: the engine-independent uniformity check.
- `results/minweight.txt`, `results/minweight_k11.txt`, `results/minweight_exact.txt`, `results/nq70_graph_adjacency.txt`, `results/minweight_scan.txt`: the stabilizer minimum-weight searches (IBM's instance and the generated ones) and the graph they ran on.
- `results/hw_marginals.txt`, `results/rng_check.txt`, `results/rng_scan.txt`, `results/rngbits_seeds_*.txt`, `results/draw_check.txt`: the low-order marginal test of the shots and samples, the random-stream identity check and scan, and the drawing-step test.
- `results/minweight_case1_k11.txt`, `results/minweight_split_test.txt`, `results/nq70_graph_geninverse.txt`, `results/nq70_graph_kernel.txt`: the second enumeration of the minimum-weight proof, its validation on smaller instances, and the generalised inverse and kernel it used.
- `results/fidelity_mle_synthetic.txt`: the fidelity estimators on synthetic shots of known fidelity.
- `results/ceiling_tightness.jsonl`, `results/ceiling_tightness.cmd.txt`: widths of random graphs and brickwork instances against the ceiling, and the command that produced them.
- `results/frontier_instances/`: order searches for the 74-, 76-, 78- and 80-qubit generated instances.
- `results/frontier.jsonl`: the frontier search output.
- `README.md`: the test table, the commands that reproduce each number, and the modes of the samplers.